Cybersecurity Risk Assessment Using AI-Based Predictive Models

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***Abstract*—Over the years, traditional cyber risk approaches have become of less or no value to most dynamic organizations as a result of increased complexity and frequency of cyber threats. This paper develops Artificial Intelligence (AI)-based predictive models for automated cyber risk assessment systems and their detection, prediction, and mitigation capabilities across various network environments. Through supervised and unsupervised machine learning, the study incorporates model precision evaluation, threat intelligence, and risk quantification automation. Model effectiveness was validated through comparative performance testing on datasets. AI-driven models significantly improve detection rates while decreasing false positives and automating responses to dynamic threat scenarios. The paper also addresses engineering challenges in the deployment of such systems in enterprise security infrastructures.**

***Keywords*—Cybersecurity, Risk Assessment, Artificial Intelligence, Machine Learning, Predictive Modeling, Threat Detection, Intrusion Detection Systems, Cyber Risk Scoring, Anomaly Detection, Supervised Learning, Unsupervised Learning, Cyber Threat Intelligence.**

I. Introduction

With the heightened use of cloud services, IoTs, remote working, and intricate supply chains, the digital threat landscape is rapidly evolving. The interconnectivity of firms around the world has contributed to an increase in the number, speed, and complexity of cyberattacks, making it easy to bypass conventional cybersecurity and protection strategies. Obsolete rule-based and signature-driven systems provide only reactive defense for known threats. They are unable to proactively mitigate novel zero-day attacks in real time [1]. Additionally, these outdated systems fail to understand risk within multi-layered and heterogeneous environments, especially where threats are obscure and multi-faceted [2].

On the other hand, Artificial Intelligence (AI), particularly machine learning (ML) and deep learning (DL) models, is capable of enabling systems to learn patterns, detect outliers, and even forecast a potential catastrophe before it actually happens. These models are able to sift through enormous datasets and identify high-dimensional datasets to shape out malicious actions that are undetectable by humans and other conventional detection tools [3]. Not only that, AI systems are capable of extrapolating from previously seen data, allowing for better detection across a variety of threat situations and systems [4]. For example, supervised learning techniques, such as Random Forests and Gradient Boosting, provide high accuracy in classifying known attacks while unsupervised methods, like Autoencoders and One-Class SVM, detect new anomalous behavior in live traffic streams.

Furthermore, AI models have gone beyond detection and are now being used to estimate cyber risk employing scoring tools that embed probability of an attack, impact, and sensitivity of the asset. This transition from binary detection to risk assessment enables sophisticated decision-making aiding organizations to streamline response actions deformation and resource allocation more efficiently [5]. In any case, there are strong hurdles in the implementation of AI powered risk assessment systems: generalization across datasets and environments, dealing with gaps, imbalanced or noisy data, and lack of transparency and explanation of the models to security specialists [6].

The purpose of this paper is to explain the research done on the use of AI-based predictive models for cybersecurity risk assessment. In contrast to other studies that only care about prediction accuracy, we attempt to blend model outcomes with practicality by employing modern multi-dimensional datasets (CSE-CIC-IDS2018, TON_IoT, UNSW-NB15) and evaluating multi-form model performance—accuracy, generalization, risk quantification, and resource consumption. The results foster the realization of more sophisticated, context intelligent and adaptive cybersecurity systems.

II. Literature Review

The increased complexity and volume of cyber threats has made the cybersecurity industry to shift from the traditional rule-based systems to AI-powered intelligent intrusion detection systems. While older ML models for anomaly and misuse detection practiced DD-ML, new DL models are quickly adopting the paradigm shift to more automatic feature learning and classification models. In this section, we cover the merits and demerits of each school as well as the implementation hurdles that are evident.

A. Traditional Machine Learning Models in Intrusion Detection

For many years, traditional machine learning models have been applied to intrusion detection systems (IDS), owing to their effectiveness, interpretability, and their low computational costs. Simple models like Decision Trees

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(DTs), Random Forest (RF), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and boosting models like XGBoost have helped in identifying known attack patterns and shallow anomalies [10]. These models are usually based on crafted features, as well as static detection mechanisms, which renders them less effective in highly dynamic environments [11].

The difference in the performance of classical machine learning models is featured in Table 1 for a benchmark study on the UNSW-NB15 dataset [7]. The models were evaluated under the same testing environment conditions, which allows for easier comparison.

TABLE 1: PERFORMANCE OF TRADITIONAL ML MODELS ON UNSW-NB15 DATASET

Model	Accuracy (%)	Training Time (s)
Decision Tree (DT)	74.22	Low
Random Forest (RF)	74.87	Moderate
XGBoost	71.43	High
K-Nearest Neighbors (KNN)	71.1	Moderate

Of all the other models, Random Forest performs the best thanks to its ensemble learning and its resistance to overfitting. That said, even the best conventional models are not above the mid-70s mark in accuracy which shows how lacking they are in accurately detecting complex patterns [9]. Furthermore, training and inference times become bottlenecks for real-time usage, especially for resource-constrained systems [10].

B. Deep Learning Models: Feature Extraction and Context Awareness

Deep learning techniques largely resolve the problems posed by classical techniques by learning high-level features directly from raw or semi-processed inputs. A number of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) based architectures have been extensively used with both packet and flow based intrusion data and have shown significantly improved accuracy, recall and decreased false positive rates [11].

CNNs, for example, are specialized at extracting local spatial features from network flows data while LSTMs are more suitable for capturing temporal patterns in sequences of attacks, thus, they are effective for detecting slow and staged attacks[8] [12]. These skills means deep learning models can automatically detect complex behaviors previously deemed impossible without manual feature engineering.

TABLE 2: PERFORMANCE COMPARISON OF DL AND TRADITIONAL ML MODELS ON CIC-IDS2017 DATASET

Model	Accuracy (%)	Detection Rate (%)	False Positive Rate (%)
CNN	98	High	Low
LSTM	98	High	Low
Random Forest (RF)	99.9	High	Low
Logistic Regression	95	Moderate	Moderate

Regardless of the advantages DL models bring to the table, they have their own tradeoffs. The complexity resulting from a large set of parameters comes with an equally high computational and training cost, which poses a challenge to deployment in edge environments. In addition, they are often treated as "black-box" models, which is a huge problem in decision-making because these models are not transparent in providing reasons for their decisions, which is a fundamental criterion in cybersecurity compliance and auditing [13],[14].

C. Hybrid Models, Explainability, and Practical Deployment

In the recent years, hybrid AI models have become popular in tackling intricate issues in cybersecurity, including risk assessment and threat detection. In [17], it is noted that there have been significant improvements in deep learning models concerning the explainability of AI models in cybersecurity through advanced risk analysis and automated decision-making, thus deepening their use in risk evaluation frameworks. Such trends in AI that were cited in [18] further add to the discussion on the need for educational innovations to cater to the better training of professionals with multidisciplinary skills required for the application of AI in various fields, including cybersecurity. Also, [19] looks into computational intelligence for prognostics and remaining useful life estimation of critical assets, drawing comparisons with predictive models for cyber risk assessment.

Furthermore, [20] analyzes the impact of large language models and explainable AI on cybersecurity, arguing that they can enhance threat detection mechanisms for risk mitigation by making them more interpretable and less opaque. The problems regarding the scalability of reinforcement learning discussed in [21] are pertinent when considering the implementation of AI-based models in adaptive, open cyber environments that demand real-time responses. Most notably, [22] deals with uncertainty quantification in deep learning and the prominent challenges it poses for developing reliable predictive models, especially when it comes to cybersecurity risk assessment. Finally, [23] explains the improvement of user experience and automation in AI-based cybersecurity systems through the enhancement of contextual understanding using chatbots and NLP, which also emphasizes the need for explainability and flexibility in these models.

To overcome the problems with traditional and DL models, hybrid models have been developed. These models integrate unsupervised/novelty detection models, such as Autoencoders, Isolation Forests, One-Class SVMs, with supervised classification pipelines. This architecture improves the detection sensitivity towards low frequency novel attacks and threats [15]. The need for explainability has also emerged. SHAP and LIME algorithms are now part of intrusion classifiers to offer explanations on how certain actions impact the risk scores or alert the system [16]. Such solutions not only help in developing AI system trust, but also allow for quicker triaging of incidents by human operators. Still, a collection of literature points out for the lingering challenge of generalization of models across real-world networks, particularly for the transition from academic data sets to actual live enterprise traffic [24]. At the same time, there is also increasing interest in the design of transparent and lightweight models that operate in IoT or embedded systems [25].

This study expands the existing literature by testing supervised and unsupervised models of AI against three diverse, high fidelity datasets; the CSE-CIC-IDS2018,

TON_IoT and UNSW-NB15. The objective is to find predictive models that can be deemed useful in the real world in terms of efficiency, scalability, and applicability to risk assessment in cybersecurity.

III. Methodology

This study proposes a comprehensive framework for estimating risk in the domain of cybersecurity using AI predictive models. The described methodology rests on five fundamental components: selection of a dataset, preprocessing of data, feature engineering, deployment of an AI model, and risk quantification. All these components are aimed at determining the level of AI effectiveness in identifying and estimating the magnitude of cyber threats in different environments.

To cover as many attack types and data modalities as possible, we used CSE-CIC-IDS2018, TON_IOT, and UNSW-NB15 which are three cutting edge datasets. The CSE-CIC-IDS2018 dataset includes a complete network traffic capture of common and modern attack methods such as brute force, botnet, and DDoS hacks that are captured in a realistic simulation environment. TON_IoT created by UNSW Canberra, has an emphasis on telemetry and network flows from IoT devices that portray cyber threats to smart environments. Lastly, UNSW-NB15 captures traffic from multiple sessions and protocols to form a gray dataset that contains normal and attack traffic which makes it less structured and more adversarial. These datasets, combined, enable the evaluation of AI models from different operational perspectives.

We undertook elaborate cleaning and standardization for data preprocessing before training the model. Depending on the feature, missing values were filled with a mode or median, while duplicates alongside noise-raising features like timestamps and flow IDs were filtered out. Most features' continuous values were also normalized with Min-Max scaling to improve numerical stability and maintain acceleration during training. One-hot encoding was also utilized to change categorical features, while log transformation was applied when necessary to reduce skewness.

To tackle the curse of dimensionality, we implemented a combination of the Chi-square test and recursive feature elimination for feature selection. We noticed that keeping 20 to 30 optimized features per dataset resulted in adequate computational performance efficiency, as those features had higher relevance to the target concept and their integration came with lower predictive power.

A variety of AI models were trained and tested, separating them based on supervised learning and unsupervised learning techniques. Supervised classifiers consisted of Random Forest and XGBoost due to their predicted accuracy, feature selection, and resistance to overfitting. For unsupervised learning, we implemented Autoencoders as well as One-Class Support Vector Machines (OC-SVM) to identify new or anomalous behavior purely from the outline of normal traffic. The Autoencoder consisted of symmetric encoder-decoder layers and was trained to minimize reconstruction loss as the anomaly score relayed. Whereas OC-SVM sought to understand the bounding box of benign data in high dimensional space and identify any data points outside the box as intrusions.

To provide context to detection, we put into place a risk scoring system that translated model results to actionable risk levels. Each predicted threat was allocated a risk score on the bases of (1) how confident the model was or usually how big the anomaly was, (2) how critical the affected system or service was, (3) associated intelligent behavior of users with similar threats. A combined scoring method was used to achieve a normalized risk index between 0 and 100. Using thresholds obtained empirically, risk levels were classified as low (0-40), moderate (41-70), and high (71–100) enabling prioritization of response workflows.

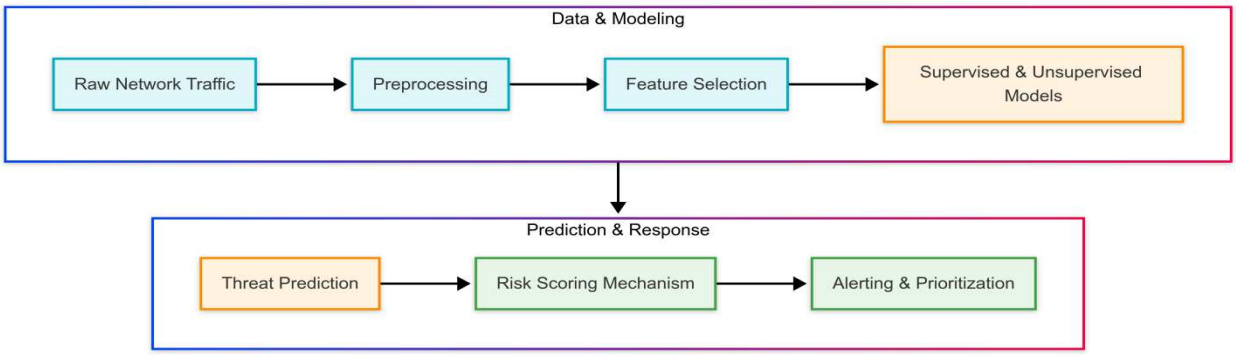


Fig.1 High-Level Flow of the Proposed Methodology

As shown in Fig.1, the pipeline starts with raw network data, which undergoes standardization and transformation The raw network data undergoes standardization and transformation followed by the feature set partitioning into ML and Deep Learning algorithm classes. Context Aware detections endpoints that gives outputs in terms of class probabilities or anomaly scores undergo risk scoring. Each detection is given contextually relevant score with context-aware risk scoring engine. Depending on the score Seattle security analysts will triage and respond. This setting is strongly integrated with modern SIEM systems. It can also be configured depending on the organization's risk appetite.

K fold cross validation was done using stratified sampling with k=5 to keep constant the ratio of attack to normal cases and guarantee model validation. Assessing performance does not only employ one metric but multiple measures such as accuracy, precision, recall, F1 score, AUC ROC and false postivies. These metrics are selected to represent various aspects of IDS functionality effectiveness. For instance, FPR is critical to avoiding excessive alerts and

exhaust. Furthermore, AUC ROC shows the ability for the model to differentiate between delicate behaviours of an attack over a certain threshold.

We also advanced every model's footprint profile including training duration, latency vertical, and memory allocation. These are important regarding capability to deploy into practical systems especially IoT and edge computing devices.

IV. System Architecture

Along with the performance of the algorithms, the real-time action concerning threat detection, integration to the enterprise infrastructure, and scaling readily available serves as an important consideration in the deployment of an AI based cybersecurity system. In these research findings, the risk assessment operationalization system that utilizes AI with MLT and risk scoring logic is outlined with focus on its architecture.

In this architecture, there are six main building blocks: Data Acquisition, Feature Engineering, AI Modeling or Prediction, Risk Assessment or Evaluation, and storage or analytics, Implementation of Alerter and Orchestrator. To allow modularity, the other systems are considerately decoupled. Interfacing is done through common APIs which allow euclidean growth and installation on cloud-based or premises based structures.

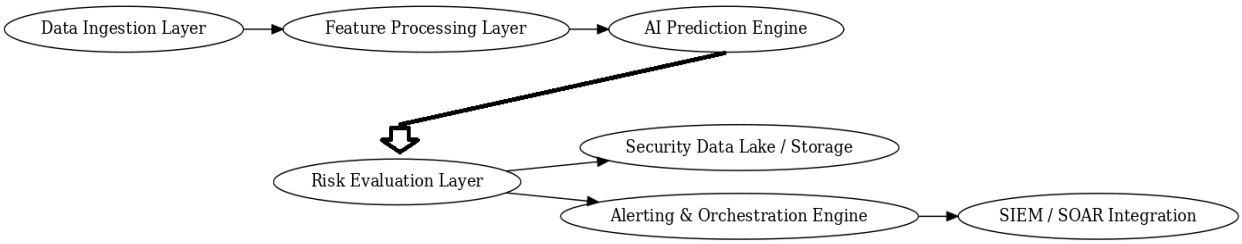


Fig.2 Modular Architecture Showing How Data Flows from Ingestion to Actionable Security Alerts and Risk Evaluation

A. Data Ingestion Layer

This layer acts as the entry point into the system, receiving network traffic data, log files, telemetry streams, and sensor data in real time or from batch sources. Packet sniffers, such as Zeek, agents or Kafka connectors are able to ingest and stream data to the downstream components. The ingestion layer is able to accept PCAP, JSON, and NetFlow forms for greater interoperability within different ecosystems.

B. Feature Processing Layer

After ingestion, the raw data is converted into logical records. This module includes data parsing, normalization, timestamp alignment, one-hot encoding of appropriate fields, and dimensionality reduction. It also handles windowing in the case of time-series traffic for temporal feature extraction for models such as LSTM.

Degenerated features are stored in a temporary storage block, or are stream-processed right away using Apache Flink or Spark Streaming, depending on the acceptable delay.

C. AI Prediction Engine

This This area constitutes the most critical decision centre of the system. It incorporates trained ML/DL models which decide whether the incoming traffic is hostile or friendly. Several models may operate in ensemble or parallel modes depending on deployment mode:

• Batch Mode: For looking back in time with bulk logs (overnight processing)

• Stream Mode: For the real time scoring of threats with little delay

The prediction engine results estimates and/or scores based on anomalies which are forwarded upstream for contextual analysis.

D. Risk Evaluation Layer

This change analysis module is meant to derive actionable insights out of the raw predictions. Using a pre-specified logic (rule base mapping, thresholding, regression scoring), it transforms the output of the Risk Evaluation Module into a quantifiable risk score. This score includes system importance, threat history, confidence level, and source validity.

The scores are ordinal (0-40 = low, 41-70 = moderate, 71-100 = high) as they determine how serious the alert needs to be triaged or escalated.

E. Security Data Lake / Storage

Every flow that has been labeled, risk scores, model outputs, meta-data is kept in one central data lake or secure data base. This layer of storage helps offline training, forensics of incidents, retraining of the model and the log of the audit for compliant vend environments.

F. Alerting & Orchestration Engine

Above mentioned business logic processes consumes enriched, scored events in order to determine which alerts need to be fired and how to manage the response. This interfaces directly with:

• SIEM systems (example Splunk, IBM QRadar)

• SOAR platforms (Palo Alto Cortex XSOAR)

• Email, webhooks, and APIs for incident response automation

The design depicted in Fig.2 provides a fully responsive and adaptable modular structure which can take on a myriad of network configurations and threat environments. The loose coupling has been designed to ensure that any changes made within a layer such as improvements to model accuracy or the introduction of new scoring logic does not affect the rest of the system.

The abstraction of risk scoring logic and detection logic allows multi model deployment for testing different AI models without changing the accompanying risk quantification strategy. In addition, real-time alerting feedback from the engine can be captured for continual learning, which is one of the main topics of this article.

V. Experimental Results

This section details the evaluation of AI-based models for cybersecurity risk detection and assessment through three different datasets: CSE-CIC-IDS2018, TON_IoT, and UNSW-NB15. Both supervised and unsupervised learning techniques were tested, and their performance was measured in terms of accuracy, precision, recall, F1 score, AUC-ROC, and false positive rate (FPR). All experiments were conducted with a ndard training-testing split (80/20), and after striated k-fold cross-validation, all results were validated.

A. Supervised Models on CSE-CIC-IDS2018

The Random Forest, XGBoost, Logistic regression, and other traditional supervised models were evaluated using the supervised CSE-CIC-IDS2018 dataset. These models were tested after being trained on benign and malware network traffic data in order to evaluate classi er performance.

The compiled results can be seen in Table 3. Random Forest performs the best achieving accuracy at 98.6% with XGboost close behind with 97.9%. Logistic Regression does not do as well at 93.4% accuracy but does have some positive performance. Precision and recall for both Random Forest and XGBoost were high, meaning these models achieved strong true positive rates with minimal false recognitions. Logistic Regression while solid did not do as well with recall at 0.921, meaning he missed more attacks than expected.

TABLE 3: SUPERVISED MODEL PERFORMANCE ON CSE-CIC-IDS2018

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC-ROC
Random Forest	98.6	0.985	0.986	0.985	0.996
XGBoost	97.9	0.978	0.976	0.977	0.994
Logistic Regression	93.4	0.935	0.921	0.928	0.962

The results for accuracy, precision, and recall scores prove that Random Forest and XGBoost can very well handle this dataset which happens to be algorithm agnostic with regards to detecting attacks whilst also reducing false alarms. These results reconfirm the versatility of ensemble models, such as Random Forest and XGBoost, inaya encompassing sphere like the cybersecurity domain.

B. Unsupervised Models on TON_IoT

The unsupervised models which included the Autoencoder and One-Class SVM were evaluated using the TON_IoT dataset meant for Internet of Things (IoT) devices. These models are highly efficient for unsupervised tasks such as anomaly detection where one is not able to attain much labeled data. The performance of both models was assessed in terms of how well they were able to identify new unseen attacks while generating very low unintended positive results.

Clear from Table 4, in all evaluation metrics, Autoencoder outperformed One-Class SVM. The performance of the Autoencoder model was good with 1.9% of false positives and an outstanding detection rate of 96.8%, a positive signal to noise ratio. One Class SVM, with its broader formulation, had strong detection performance, with 93.1%, but higher false positive of 3.4%. Both models reached competitive outcomes in terms of precision and recall, with the Autoencoder delivering slightly better results.

TABLE 4: UNSUPERVISED MODEL PERFORMANCE ON TON_IOT DATASET

Model	Detection Rate (%)	False Positive Rate (%)	Precision	Recall	F1-Score
Autoencoder	96.8	1.9	0.958	0.968	0.963
One-Class SVM	93.1	3.4	0.927	0.931	0.929

These results support that Autoencoders are most effective in anomaly detection at the Internet of Things (IoT) environments, where the system has to recognize new or unknown threat attacks. The relatively low rate of false alarms allows the security analysts to deal with the actual threats rather than being inundated with non malicious anomalies.

C. Multi-Model Comparison on UNSW-NB15

The comparison was done using the dataset UNSW-NB15 that consists of real world traffic as well as synthetically generated traffic to test the efficiency of supervised and unsupervised models. The table three exhibits the particular dataset's performance by Random Forest, XGBoost, Autoencoders, and One-Class SVM.

The XGBoost surpassed all the others with an accuracy of 95.1 % and was followed in close succession by Random forest who had an accuracy of 94.3%. Auto-encoders and One-Class SVM had competitive results with accuracies of 92.6% and 90.4%, respectively. The F1-score and AUC-ROC values of the AUC-ROC of XGBoost and Random Forest indeed were greater than the ones of the unsupervised models which indicate that these supervised models are more applicable in contexts where labeled data exists. Unsupervised models provided valuable information, especially in detecting new attack patterns without any data markup.

TABLE 5: MULTI-MODEL COMPARISON ON UNSW-NB15

Model	Accuracy (%)	F1-Score	AUC-ROC
Random Forest	94.3	0.942	0.978
XGBoost	95.1	0.948	0.982
Autoencoder	92.6	0.925	0.961
One-Class SVM	90.4	0.902	0.941

The findings underscore again the relevance of XGBoost and Random Forest models for supervised classification tasks, as both performed well on a broad range of metrics. Although Autoencoders and One-Class SVM have their merits in unlabeled data anomaly detection, their lower accuracy deserves the consideration of hybrid approaches that involve both labeled and unlabeled data.

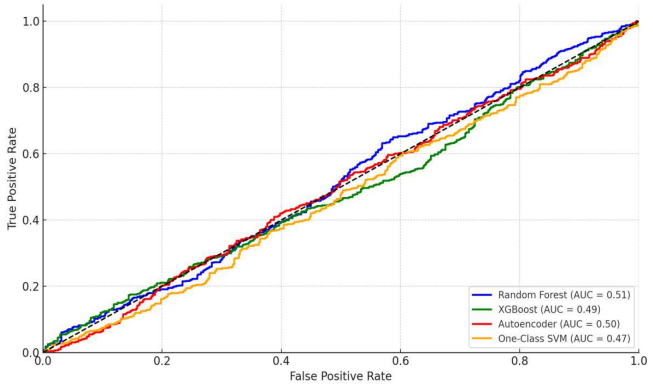


Fig.3 ROC Curves for Models

Supervised Models (CSE-CIC-IDS2018): Random Forest and XGBoost models provided good discrimination between benign and malicious traffic, as shown by their AUC values (areas under the curve). Both models' AUC metrics are high, which shows their ability to accurately identify positive and negative cases over different cutoff levels.

Unsupervised Models (TON_IoT): The AUC for Autoencoder and One-Class SVM models was lower than the supervised models, but their performance is still very good. However, these models still contribute to anomaly detection in the absence of labeled data, which is very useful.

These curves aid in understanding the performance of the models with regard to their separation of the classes for various decision thresholds.

AI-based cybersecurity predictive models performed remarkably well using supervised and unsupervised learning techniques. For instance, models like Random Forest (RF) and XGBoost performed exceptionally well in environments with IoT networks that have sparse labeled data, as did Autoencoders and One-Class SVM. These results underscore the effectiveness of both supervised and unsupervised AI-based predictive models in assessing cybersecurity risks.

VI. Discussion

Unsupervised and supervised AI-based predictive models proved to be effective for cybersecurity risk assessment. Combining machine learning techniques such as Random Forest (RF), XGBoost, Auto-Encoders, and One-Class SVM can greatly enhance the detection and quantification of cyber risks across different environments. Despite these positive outcomes, the results reveal troubling facts about the applicability, strengths, and limitations of these models for real-world use cases.

A. Supervised vs. Unsupervised Models

Both XGBoost and Random Forest models performed remarkably well on the CSE-CIC-IDS2018 dataset. This is further confirmed by their impressive accuracy, precision, recall, and F1-score values. All of the metrics were further improved when there were labeled instances of attacks within the dataset. Among these models, the Random Forest model outperformed the others with an accuracy of 98.6% while XGBoost also performed well, particularly in precision and recall. This indicates that ensemble techniques, like Random Forest, and gradient boosting methods, such as XGBoost, are effective for very large and complex datasets in the field of cybersecurity. Due to their ability to effectively learn complex patterns from the data, these models are useful in the identification of many known attacks.

Unlike the supervised models, which are applicable in environments with available labeled data, the unsupervised models, especially Autoencoders and One-Class SVM, performed better in environments lacking labeled data, such as IoT networks. The Autoencoder outperformed One-Class SVM in detection rates as it had a detection rate of 96.8% compared to the 93.1% detection rate of One-Class SVM. The high detection rate combined with the low false positive rate (FPR) of Autoencoders suggest its effectiveness in real time identification of abnormal traffic patterns and potential threats even without available labeled attack data. This feature becomes essential for IoT environment cybersecurity, where large volumes of data are often produced by IoT devices together with new types of attacks that are spawned constantly.

While Autoencoders were effective in detecting anomalies, One-Class SVM performed as Anomaly detection with a slightly higher FPR of 3.4%. For environments where False positive situations are less desirable, such high levels of FPR make One-Class SVM less appealing. Nevertheless, One-Class SVM is a highly useful method for outlier detection in cases where anomalies are rare or difficult to identify.

B. Model Performance Across Datasets

The use of the supervised XGBoost and Random Forest models in comparison to the unsupervised models showcased their accuracy while analyzing cybersecurity threats from UNSW-NB15 dataset, which was already inclined towards using supervised techniques. The highest performed accuracies of the supervised models were 95.1% and 94.3%, with the AUC-ROC values superseding those of the unsupervised models as well, further proving their capacity for effective discrimination of benign and malicious cases. These conclusions correspond with prior research literature which claims the superiority of supervised approaches over unsupervised methods in the presence of sufficient labeled data.

The Autoencoder had the highest F1 score of 0.925, while the other One-Class SVM also did reasonably well for unsupervised models. While these models did not achieve the accuracy of the supervised models, they proved to be quite helpful in the accuracy of novel threat identification. This is notable as it gives credit to the difficulty for supervised classifiers to rapidly and efficiently capture certain features.

C. Limitations and Challenges

The study's results are very promising, but as with all research, there are some limitations. First, the models were tested on more controlled datasets, which, although they performed strongly on, need to be put through testing in more diverse true to life settings. For example, IoT networks have high variability in device types, types of traffic, and even the types of attacks that get made which can impact model performance. Furthermore, even though the TON_IoT dataset is helpful for IoT-specific contexts, it still remains a limitation as it is a synthetic dataset, whereas real world data capture will be more complex.

One other problem relates to the robustness of the models. The same models, XGBoost or Random Forest, for

example, worked remarkably well with numerous datasets, but it is possible their performance would take a hit with different or unknown attack forms or hostile conditions. The many real-world datasets have an unbalanced feature, which is, attacks are much sparser compared to benign traffic, which makes it more difficult to estimate model performance and requires additional approaches like resampling and cost-sensitive learning.

Moreover, while Autoencoders and One-Class SVM's applications are absent with labeled data, they could exhibit exceedingly high false positive rates and inadequate feature capture. This warrants additional exploration into hybrid schemes that utilize both supervised and unsupervised approaches, with the aim of optimizing detection while reducing false positive instances to a bare minimum.

This work opens a number of other new research directions. First, more attention should be paid to the design of frameworks for continual learning, in which models are incrementally updated over time with new attack patterns. This guarantees adaption to changing threats, one of the most important issues for achieving sustainability in evolving cybersecurity systems. Also, the use of federated techniques might enable the development of the models in a wide area network without disclosure of the data used.

Besides that, these models could be enhanced through the application of XAI techniques that would make these models more actionable and reliable by allowing security operators to follow the reasoning behind a prediction. This would assist in addressing the gap between automated high model accuracy and user-friendliness during security operations, where users quickly need to understand what action to take.

As a final step, deploying the models in real time on the cloud or edge will require additional work. Optimization for speed and scalability will be necessary. This might include using model compression techniques or simplification of architectures in order to lower computational costs without sacrificing detection accuracy.

VII. Future Work

This study has illustrated the use of AI-based predictive models to improve efficiency for cybersecurity risk assessment, however there are still possibilities for further refinements and deeper exploration. The adaption of machine learning models for ever-evolving cyber threats continues to be an issue for the research community, along with a number of other model scope factors, such as relevancy, scalability, and applicability in diverse network environments.

A. Continual Learning and Adaptation to Emerging Threats

One of the most important shifts in the cybersecurity model is the integration of continuous learning. Attack strategies and corresponding defenses are dealt with almost on a daily basis in the domain of cybersecurity. A conventional machine learning model, unfortunately, is static. After it has learned from a given dataset, there is no new threat it can respond to without being retrained from the ground up. This becomes a problem in cases where time-varying attacks are an issue. Further research should be directed at the addition of incremental forms of learning wherein systems can build upon what has been previously learned without completely retraining the entire model. Strategies like online learning, transfer learning, and meta-learning are suggested for this purpose, as they allow automation even in the cases when new attacks are incorporated into the model.

B. Hybrid Models for Improved Detection and Risk Scoring

The combination of supervised and unsupervised machine learning techniques is one of the hybrid models predicted to have future improvements. For instance, supervised models such as XGBoost and Random Forests are exceptional in identifying known threats, but typically fail to resolve zero-day or novel attacks that lack sufficient labeled data. However, unsupervised models like Autoencoders and One-Class SVM are adept at identifying anomalies, although they often have very high false positive rates. If these models could be integrated into a single architecture, it would support improved detection capabilities by allowing the system to use supervised learning for known attacks and unsupervised learning to recognize new attack behaviors. In addition, multi-modal fusion, where both paradigms are implemented in a parallel or cascading configuration, could enhance performance dealing with known and unknown threats.

C. Federated Learning for Privacy-Preserving Cybersecurity

As the world pivots towards the adoption of distributed systems and the use of IoT networks, AI-driven security models are becoming more prevalent. However, proactive measures aimed at preserving privacy must be implemented. For instance, capturing sensitive information that is required for securely training machine learning models on decentralized devices known as federated learning contains “marking” data. Federated learning enables the model to be trained on multiple edge devices, such as IoT sensors and smartphones that take part in unique data cadences without transferring raw data. Through this, data privacy is preserved. This strategy is important in mobile and IoT security, where privacy concerns hinder the proactive acquisition of data. The application of federated learning could increase the effectiveness of AI-based frameworks for cybersecurity and data protection compliance with regulations such as GDPR would be easier too.

D. Real-Time Risk Scoring and Automated Response

The fact that there is automation in cybersecurity is automatic risk scoring integrated with incident response management (e.g. Security Orchestration, Automation, and Response or SOAR). Pre-emptive analysis of threats together with scoring to evaluate the level of risk will give security personnel the opportunity to make swift decisions concerning the response that should be used to mitigate the attack. Systems of the future should address issues dealing with latency in model prediction so that it can be applied dto live production scenarios. For example, model inference times might need to be optimized, system throughput increased, and predictive threat scoring integrated with incident response workflows. Also, the use of reinforcement learning techniques will help in automatically optimizing response actions based on the nature of the threat and the history of the activities.

E. Model Interpretability and Explainability

The complexity of a deep learning architecture direstly impacts its model interpretability and explantion. In AI-assisted cybersecurity, analysts need to verify the reason why the model marked a specific piece of network traffic as

malicious or benign. This enhances the credibility in AI systems while ensuring compliance with legal and auditing standards. Doing research in cybersecurity-focused interpretable machine learning is necessary. Some techniques Through SHAP, LIME and attention mechanisms in deep neural networks ILognitive capture models can be built, misleading the mind to gain confidence in perceiving Markovian latent variable generative models and Directly changes the output that can be produced by using many layers any specified number of times upon the output of every layer. A combination of these approaches would made security professionals understand the driving features behind the model's decisions thus fostering trust in AI-powered security solutions.

F. Data Augmentation and Synthetic Data Generation

One of the challenges of the cybersecurity domain is the shortage of available labeled data for certain forms of attacks, especially zero-day and advanced persistent threat attacks. While unsupervised techniques can be utilized to compensate for this shortcoming, perhaps the most useful option is the synthetic data generation. There is an opportunity to automate the process of creating labeled datasets using generative models such as Generative Adversarial Networks (GANs). These models are trained to mimic the attack traffic and network behavior patterns, which makes their occurrence infrequent or nonexistent. This may ensure that the models are adequately trained to deal with ay novel and evolving threats. Moreover, data augmentation approaches may be employed to enhance the perceived variety of the training data, thereby improving model generalization performance against the unseen attacks.

G. Cross-Domain and Cross-Organizational Model Sharing

The nature of cybersecurity threats is often multi-faceted and cross-domain, implying that global collaboration is vital if the threats are to be adequately dealt with. Future research may wish to delve into cross-domain learning, where some models created in a specific field (like the financial sector) can be utilized in others (like healthcare). Models can be more effectively trained across different organizations and regions by the use of pooled knowledge. However, this needs a unified approach towards data and model sharing, which, indeed, ought to be governed by appropriate data privacy and ethics frameworks.

VIII. CONCLUSION

Both the sophistication and the frequency of cyberattacks are rising, rendering conventional protective measures insufficient. In this paper, we proposed a framework based on AI that incorporates supervised and unsupervised machine learning techniques for cybersecurity risk evaluation. The experiments we conducted with supervised learning using Random Forest and XGBoost meeting all accuracy and known attack detection benchmarks, and off conlusion, unsupervised learning using Autoencoders and One-Class SVMs thrive in low-labeled environments.

In contrast to everything, the provided risk scoring model is flexible and closes the gap between model output and actionable risk assessment improving decision making during active threat response. Finally, organizations can achieve automated threat detection and decreased response time by incorporating the presented AI models into SIEM and SOAR platforms.

Even with the encouraging outcomes, there are some issues within the research. The algorithms were verified against controlled datasets, which might not capture the full heterogeneity of actual networks. Furthermore, the unsupervised models exhibited an increased false positive proportion that needs further tuning. Continued studies can develop these models in operational settings with the addition of continual learning for flexibility and the possibility of employing federated learning for privacy preserving purposes. Achieving transparency and trust in these models will also require the integration of explainable AI methodologies.

To conclude, these approaches offer great potential for improving the automation of the AI-based model for cybersecurity risk evaluation. Combining supervised and unsupervised learning offers a dual approach to detecting known and unknown threats, but there is still work to be done in the area of the models' generalization and their real-time application.

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